

Data visualization exercise

Download this file's source (day-3-viz-ex.qmd) and open it in RStudio, Positron, or VS Code. Work through the code and associated questions. *Render as you go to ward off problems as early as possible.*

For your own good, avoid AI tools for this task. Use only `?function_name`, the `ggplot2` reference, and any other R documentation.

Setup (given)

```
library(tidyverse)

counties <- left_join(
  read_csv("https://princeton-ddss.github.io/r-bootcamp/files/county_data.csv"),
  read_csv("https://princeton-ddss.github.io/r-bootcamp/files/county_population.csv"),
  join_by(GEOID == fips)
)
```

1 Part 1: the county data

1.1 Rent and income distributions by group

Plot the distribution of rent and *median household income* by region using `geom_density()` and a `facet_*` function.

1.2 Publication-ready

Save the last plot as a PDF.

2 Part 2: mapping America's partisan shift

2.1 Fetching data

1. Go to the MIT Election Lab's County Presidential Election Returns 2000–2024 on the Harvard Dataverse.
2. Download `countypres_2000-2024.tab` as a `.csv` or `.tab` file.
3. Save it in a `data/` folder next to this `.qmd`.
4. Load the data and inspect it however you wish.
5. Once you understand how the table is structured, move to the next section.

2.2 Fixing problems

Are all of the data types correct? If so, skip this subsection. If not, fix them.

There are also some bookkeeping rows in a few states in 2024 (`TOTAL VOTES CAST`, `OVERVOTES`, ...). They all have an empty `party`. Get rid of them.

2.3 Republican vote share over time

Build a new data frame called `rep_votes` with one row per county per year. Add a column `rep_share` that records the Republican **two party vote share** [meaning, $R \text{ votes} / (R \text{ votes} + D \text{ votes})$] for that county–year, then pivot the data wide and add a column `shift` recording the difference between the 2024 and 2012 Republican two-party vote share for each county.

2.4 Visualizing county-level shifts

Plot 2024 share against 2012 share with a reference 45° line (representing no shift). What does a county above the 45° line mean?

2.5 Mapping the shifts

Load the county boundaries from this morning's session and map the county-level shifts. Your color scale should use red and blue properly to indicate partisan shift. Hint: Use one of these functions. (Which one?)

3 Part 3: Putting it together

3.1 Republican shifts and county-level income

Join the income data to the shape data (with its `shift` variable). Create a plot showing the relationship between county-level income and the 2012–2024 Republican vote share shift.

3.2 Thinking like a social scientist

How *might* this be a misleading analysis? Hint: Look at [Gelman et al. 2010](#), *Red State, Blue State, Rich State, Poor State*, Chapter 1. How would we examine this relationship properly?

Election data: MIT Election Data and Science Lab, "County Presidential Election Returns 2000–2024", doi:10.7910/DVN/VOQCHQ (CC0).